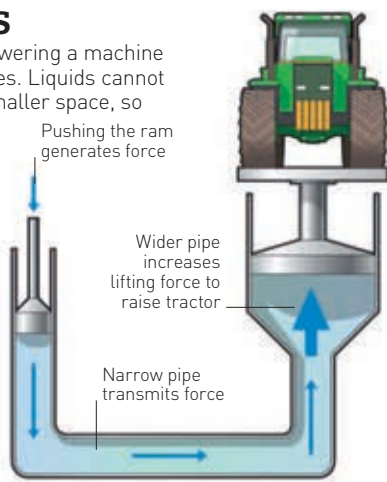


HYDRAULICS

Hydraulics means powering a machine using liquid-filled pipes. Liquids cannot be squeezed into a smaller space, so a pipe filled with oil can be used to exert force. If the pipe is wider at one end than the other, the force is increased.



HOW A HYDRAULIC RAM WORKS

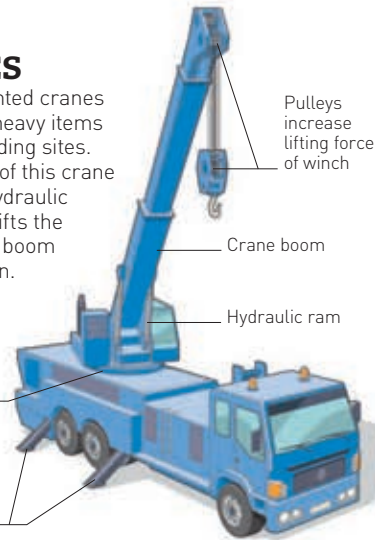
Since the lift pipe is wider than the ram pipe, the lifting force is multiplied.

CRANES

Truck-mounted cranes move very heavy items around building sites. The engine of this crane powers a hydraulic pump that lifts the main crane boom up and down.

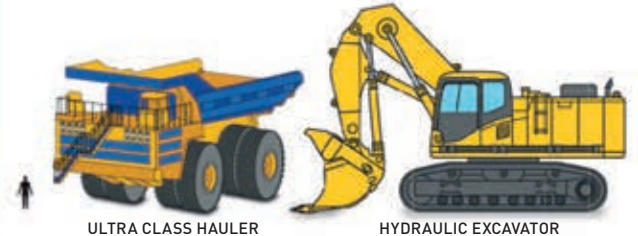
Turntable swings the boom over a large area

Hydraulic stabilizers stop the crane from tipping over



MIGHTY MACHINES

The world's biggest dump trucks – or ultra class haulers – stand at around 8 m (26 ft) high. These mechanical monsters are used in mines and can carry a staggering 500 tonnes of debris – the weight of 38 elephants. At 10 m (33 ft) high, the largest hydraulic diggers weigh around 980 tonnes. They can shovel nearly 1,000 tonnes of material an hour.



EMERGENCY VEHICLES

The most important trucks are those that save lives. Fire engines are equipped with ladders, water tanks, and other tools. Armoured SWAT trucks are used by the military and police, while ambulances ferry the sick and injured to hospital.



DIGGERS

Also known as excavators, these machines use a bucket on the end of a hinged arm (boom) to dig into the ground. Wheeled diggers are suitable for moving across hard surfaces, while tracked wheels are best for mud. Loaders are used to scoop up loose material, such as gravel, from the ground.



SKILLED OPERATORS USE BACKHOE LOADERS TO PERFORM STUNTS CALLED "DIGGER DANCING"